



Winners and losers of generative AI: Early Evidence of Shifts in Freelancer Demand

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ABSTRACT

We examine how ChatGPT has changed the demand for freelancers in jobs where generative AI tools can act as substitutes or complements to human labor. Using BERTopic we partition job postings from a leading online freelancing platform into 116 fine-grained skill clusters and with GPT-4o we classify them as substitutable, complementary or unaffected by LLMs. Our analysis reveals that labor demand increased after the launch of ChatGPT, but only in skill clusters that were complementary to or unaffected by the AI tool. In contrast, demand for substitutable skills, such as writing and translation, decreased by 20–50% relative to the counterfactual trend, with the sharpest decline observed for short-term (1–3 week) jobs. Within complementary skill clusters, the results are mixed: demand for machine learning programming grew by 24%, and demand for AI-powered chatbot development nearly tripled, while demand for novice workers declined in general. This result suggests a shift toward more specialized expertise for freelancers rather than uniform growth across all complementary areas.

1. Introduction

The impact of Large Language Models (LLMs) on labor markets remains an open question, with early evidence suggesting a wide range of outcomes. On one hand, LLMs have been shown to reduce demand for certain tasks like writing and translation, where they can substitute human labor (Demirci et al., 2023; Qiao et al., 2023; Hui et al., 2023; Liu et al., 2023). On the other hand, laboratory experiments have shown that LLMs can be used as complementary tools that enhance productivity, particularly for less experienced workers (Brynjolfsson and McAfee, 2014; Noy and Zhang, 2023). These mixed results align with the view of Autor (2024), that the effects of new technology on work depend heavily on how it is deployed by firms, governments, and other stakeholders. To better

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understand the impact of generative AI on work and employment, we examine the effects of the ChatGPT launch on labor demand across fine-grained skill clusters that can be substituted or complemented by LLMs.

While LLMs and Generative AI (Gen AI) are fairly new, there is extensive literature estimating the exposure to automation – i.e., the technical feasibility of occupations being replaced by AI – using O*NET task data on occupations and patent data. Frey and Osborne (2017) estimated that half of the US labor force could be in occupations at risk of automation. Arntz et al. (2016) offered a nuanced view of these findings through a task-level analysis and concluded that less than 10% of the jobs in OECD countries were at high risk of automation. In 2017, Brynjolfsson and Mitchell (2017) argued that AI would most likely redefine work within occupations. Webb (2019) used Natural Language Processing (NLP) to study occupation task descriptions and patent documents, suggesting that high-skill tasks are more amenable to AI replacement, potentially decreasing wage inequality between the highest and lowest earners without impacting the top 1% of earners. Albanesi et al. (2023) highlighted a positive association between AI exposure scores and employment growth across 16 European countries from 2011 to 2019.

Several empirical studies have explored the impact of recent technological innovations on the labor market. Autor et al. (2024) found that in recent decades, labor demand-eroding effects of technologies intensified while demand-increasing effects did not. Using online platform data, Stephany and Teutloff (2024) documented that workers with machine learning and AI skills can earn a premium compared to workers who have otherwise similar skill sets but lack AI proficiency. For online freelancers, Lysyakov and Viswanathan (2023) found that in response to the threat of image-generating AI, designers on an online crowd-sourcing platform split into two groups: low-tier designers exited the market, while higher-capability designers moved away from direct competition with AI by focusing on more complex tasks.

Regarding LLMs, a small but rapidly growing literature examines their labor market impacts. Eloundou et al. (2023) estimated that approximately 80% of the U.S. workforce could see at least 10% of their job tasks influenced by the deployment of (LLMs), with around 19% of employees potentially experiencing an impact on at least half of their tasks. Noy and Zhang (2023) conducted an experiment and found that ChatGPT increased the productivity of mid-level professional writing tasks. Similarly, Brynjolfsson et al. (2023) found that access to a generative AI tool led to a 14% increase in productivity, as indicated by the number of issues resolved per hour. Novice and low-skilled workers experienced the highest increase in productivity, while the impact on seasoned, skilled workers was negligible. Experimental evidence also showed that LLM-based code writing assistants such as GitHub Copilot increased developer speed by over 50% (Peng et al., 2023), with less experienced programmers benefiting the most.

Outside laboratory settings, initial evidence shows the short-term effects of LLMs on labor markets from online labor platforms. On the demand side, generative AI significantly reduces the number of freelance projects in automation-prone occupations (Demirci et al., 2023). Demirci et al. (2023) used a skill co-occurrence network approach to create 15 clusters of jobs that fall within three broad groups: manual-intensive, writing automation-prone, and image-generation jobs. They find that writing automation-prone and image-generation jobs decrease compared to manual-intensive jobs. On the supply side, effects vary between workers: those in occupations highly exposed to generative AI substitution suffer reductions in earnings and employment opportunities (Hui et al., 2023; Liu et al., 2023). However, while top-performing freelancers might be disproportionately affected by generative AI (Hui et al., 2023), those who integrate generative AI into their work attract more jobs (Liu et al., 2023). Freelancers in web development whose jobs are complemented by AI may increase their earnings, while translators face fewer opportunities and lower wages (Qiao et al., 2023).

Both experimental studies and those based on online labor platforms offer initial insights into how LLMs may affect the labor market. However, these studies have either focused on controlled laboratory settings, which may not translate to complex labor markets, or on a limited group of broad skills, potentially obscuring heterogeneous effects. Moreover, online labor platform studies have primarily examined substitution effects, while LLMs often serve complementary roles, enhancing certain tasks without fully replacing entire jobs (Eloundou et al., 2023). Given that the effects of new technologies depend significantly on their implementation and the specific skills they impact (Autor, 2024), we need a deeper understanding of the impact across skill heterogeneity and to distinguish between substitutable and complementary skills. However, conducting such a study poses two main challenges. First, skill categories on online labor markets evolve over time, complicating the use of co-occurrence networks at a fine-grained level. Second, it is manually infeasible to categorize the impact of LLMs on millions of job postings into substitutable, complementary, or unaffected categories.

NLP tools, which are frequently used in economic research, can help us overcome these challenges. As Ash and Hansen (2023) pointed out, text algorithms can solve several common problems in applied economics, including document similarity measurement and concept detection. For example, Hoberg and Phillips (2016) quantifies the relatedness of firms based on the similarity of their product offer descriptions. Hansen et al. (2023) and Adams-Prassl et al. (2020) use a language-processing framework to determine whether job postings refer to hybrid or fully remote work. del Rio-Chanona et al. (2023) use topic modeling to study how the prevalence of topics across the work discourse evolved during the Covid-19 pandemic and the Great Resignation. Most recently, LLMs have been shown to perform near the human level on a range of qualitative data analysis tasks, such as e.g. text annotation, interview analysis, or automated lexicography (Karjus, 2023). Following this line of work, Eloundou et al. (2023) used ChatGPT to quantify the extent to which occupations' work activities can be performed by LLMs.

To tackle the first challenge of evolving skill categories, we use a topic modeling approach to group job postings into skill clusters with similar semantic meanings. This allows us to include more than 3 million job postings into our analysis and identify 116 distinct skill clusters. To overcome the second challenge of assigning jobs AI exposure labels, we build upon recent LLM prompt engineering research (Eloundou et al., 2023; OpenAI, 2023; Wei et al., 2022) to classify the exposure of the skill clusters to LLMs into substitutable, complementary or unaffected groups. We identify 71 fine-grained skill clusters to be substitutable or complementary, and 45 to be unaffected. This classification allows us to move beyond existing research that manually defines writing-related services

to be the main job types affected by the launch of the ChatGPT (Hui et al., 2023; Qiao et al., 2023; Liu et al., 2023) and instead allows us to measure the impact across a broader range of substitutable and complementary skill clusters. Finally, we employ a difference-in-differences econometric model using the unaffected clusters as a control group to quantify changes in demand across substitutable and complementary skill clusters after the release of ChatGPT.

Our results show that while the demand for unaffected and complementary clusters increased after the introduction of ChatGPT, the number of job postings in the substitutable clusters decreased by 7%. The difference-in-differences analysis reveals that, compared to the unaffected clusters, substitutable clusters experienced a 25% decline, while complementary clusters as a whole showed no significant effects. When distinguishing projects by their duration, we find that the decrease in substitutable job postings was concentrated in short-term (1–3 weeks) jobs. In terms of worker experience, heterogeneity across substitutable clusters is limited. However, we find a decline in job postings for novice workers in complementary skill clusters.

At a cluster level, those related to writing for Real Estate and ‘About Us’ pages showed the most substantial decreases, at 52% and 59%, respectively. Translation jobs also saw significant impacts, with the ‘Western European Languages’ category experiencing a 23% decline. While complementary clusters did not show significant aggregate effects, demand changes were mixed within these clusters. Notably, the demand for jobs in the ‘AI-powered chatbots’ cluster nearly tripled, with a 179% increase, and the ‘Machine Learning’ cluster saw a significant rise, with demand increasing by 24%.

Our findings present a nuanced view of the impact of ChatGPT on labor demand. While we observe similar trends to those reported by Demirci et al. (2023), Hui et al. (2023), Liu et al. (2023), Qiao et al. (2023) regarding substitutable tasks, our analysis indicates that these decreases are primarily concentrated in short-term gigs. Moreover, although there is no aggregate decline in complementary skills, we find a reduced demand for novice workers in these clusters. This may appear to contradict the findings of Noy and Zhang (2023) and Brynjolfsson et al. (2023), who found that LLM tools boost the productivity of novice workers. However, this may result from the complex dynamics of the labor market, where increased productivity of novice workers within firms may reduce the need for novice freelancers. Thus, while LLMs usage enhance the efficiency of in-house novice workers, it could diminish external demand for similar freelance skills. This suggests a shift toward specialized expertise for freelancer demand rather than uniform growth across all complementary areas. Our results also show that while some complementary skill clusters experienced significant declines, others, particularly those related to programming and software development, saw substantial increases.

Overall, our analysis underscores the importance of considering the specific nature of tasks and the context of AI implementation when assessing its impact on labor markets. The heterogeneity in outcomes suggests that integrating AI into work processes can create both winners and losers, depending on how AI technologies are utilized and the specific skill sets involved.

2. Data and methods

2.1. Research design

To measure the impact of LLMs on labor demand at a granular skill level, we distinguish between skill clusters that are substitutable by, complementary to, or unaffected by LLMs. We then analyze how demand for these skills changes in an online labor platform around the time of the public release of ChatGPT on November 30, 2022.

Our analysis focuses on data from a leading online labor platform, where job postings list detailed skill requirements. We use the transformer model BERTopic (Grootendorst, 2022) to group job postings into clusters based on their skill requirements. To assess the robustness of our results, we combine hierarchical clustering with a manual inspection of all skill clusters, ensuring that job postings with similar skill requirements are clustered together. This procedure results in 116 skill clusters. The methodology of inferring skill clusters from the data is illustrated in Fig. 1.

Following LLM exposure definitions from previous literature (Qiao et al., 2023; Eloundou et al., 2023) and through prompt engineering techniques, we design prompts for labeling each skill cluster into substitutable, complementary, and unaffected with GPT-4o. We manually inspect each cluster label by examining the reasoning provided by GPT and the label coherence within hierarchical clusters. In the cases where human and GPT labels diverge, we give priority to human labels. This results in 12 skill clusters in the substitutable category, 59 in the complementary, and 45 in the unaffected category.

In the last step, similar to other studies assessing the causal impact of LLMs on the labor market (del Rio-Chanona et al., 2024; Demirci et al., 2023; Qiao et al., 2023; Hui et al., 2023), we use a difference-in-differences (DiD) model to estimate changes in demand of projects in the online labor market across treatment groups (substitution and complementarity) in comparison to the control group (unaffected).

2.2. Data

We use data from one prominent global online freelancing platform, which wished to remain anonymous. Online freelancing platforms are digital marketplaces that connect buyers and sellers of remotely deliverable work. Prominent examples are *Upwork*, *Fiverr*, or *Freelancer*. Jobs on these platforms cover a wide range of professional domains, including writing, translation, software development, design, marketing, or legal services. Employers range from individuals to startups and Fortune 500 companies (Corporaal and Lehdonvirta, 2017). When posting a job, employers specify detailed skill requirements, including a mix of broad skills such as *writing*, *graphic design* or *translation* and specific skills such as *Microsoft Word*, *Adobe Photoshop* or *Chinese English translation*. On average, projects require a median of six skills. In addition, every job posting contains a title and a description specifying the task. For the purpose of identifying skill clusters, we only consider the skill tags, not the job title or the text of the job description.

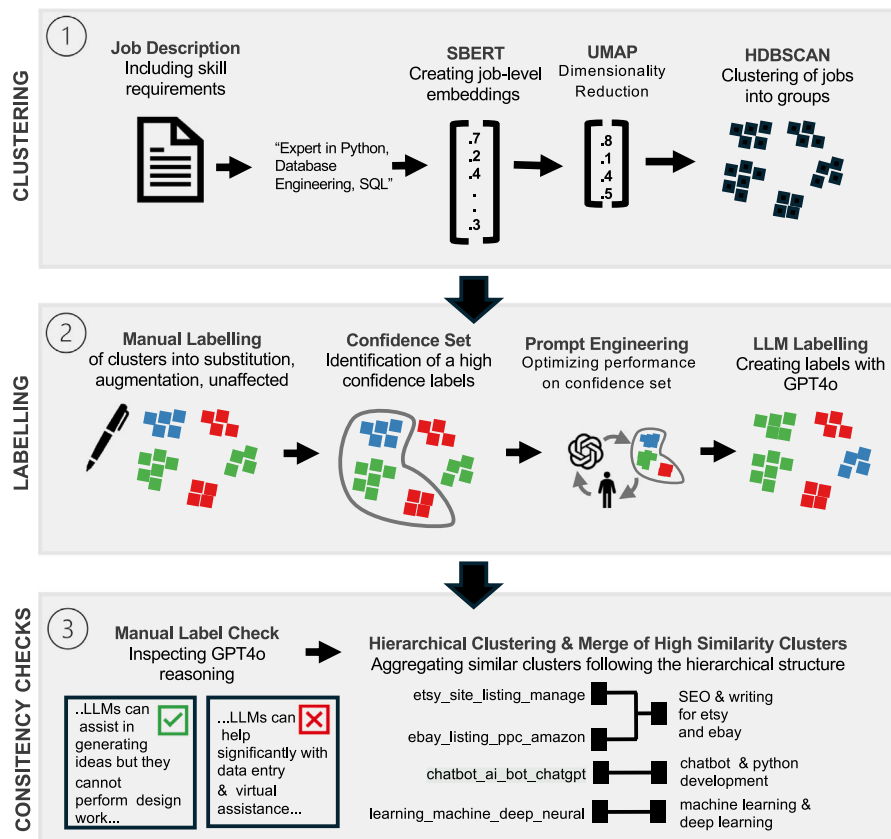


Fig. 1. Methodological overview: From job postings to AI exposure label. Our approach for creating skill clusters from job postings and assigning them to a treatment group consists of three steps. First, we use the large language model SBERT to embed the skill requirements of each job posting, creating numerical high-dimensional job-level embeddings. Through UMAP, we reduce the dimensionality and then apply the HDBSCAN clustering algorithm. Second, we manually label the resulting skill clusters into substitutable, complementary, and unaffected to create a high-confidence set that serves as a benchmark for prompt engineering GPT-4o. The best prompt is used to label all skill clusters. Third, we use GPT-4o's reasoning and the hierarchical structure of the skill clusters to verify the label consistency. Merging similar skill clusters ensures stable observations over time that are robust to the platform renaming skills.

When posting a job, the employer is initially asked to provide a title. Based on this job title, the platform automatically generates a set of skill requirements. Employers have the flexibility to edit this list of skills as they see fit. The platform curates the pool of available skills, adding new ones based on job postings and ongoing discussions with key clients (for instance, when Python2 was replaced by Python3, the relevant skill tags were updated for the new job postings. The old ones still show Python2.). The skill clusters are built in a way that the number of clusters does not change during the analysis period.

The granular information on skill demand and the fast-paced nature of this digital labor market make it an ideal setting to study the labor market implications of LLMs because changes in demand for specific skills should manifest much faster than in traditional labor market settings, making it an ideal real-world "lab" (Horton and Tambe, 2015) for observing the effects of LLMs.

Data for this study was gathered through the Online Labour Index project, as detailed by Kässi and Lehtonvirta (2018), which has been monitoring daily new job listings through the platform's API since January 2017. Further data collection on job specifics took place in multiple phases from November 2021 to August 2023. The freelancing platform, while based in the United States, operates globally, drawing a significant number of workers from countries such as India, Pakistan, the Philippines, and East Europe, as noted by Stephany et al. (2021).

We have collected information on the daily new job postings between January 2021 and September 2023, covering several million job postings. In a few cases, there were gaps in the data collection due to various software errors. These are visible as sudden "blips" in the time series of postings. Reassuringly for us, the data collection worked without errors from May 2022 to September 2023. We also demonstrate that our results remain consistent even when controlling for these fluctuations using week dummies or when excluding the few problematic time periods from our data.

Besides detailed information on the skills required for the job posting, our data includes various other metadata. In particular, we use two variables as control variables when studying effect heterogeneity: expected project duration and worker experience level. When creating a job posting, the employer is asked to provide the expected duration of the job. The duration can be one of the following: under 3 weeks, 3–9 weeks, 9–18 weeks, or 18–52 weeks. This information is not binding for either the employer or the worker; it merely reflects the employer's expectation about the contract length.